

Fault Signatures in the Spike Stream of a Closed-Loop Neuronal Culture Substrate

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Abstract—When the only feedback path of a walking humanoid is a population of cultured neurons, actuator faults must be legible from the spike stream for the loop to be diagnosable. We inject four actuator faults at $t = 2.0$ s while a Unitree G1 in MuJoCo walks at $v_x = 0.5$ m/s under closed-loop command through a 64-electrode, homeostatic, untrained rate substrate read by a fixed Nengo hub. A z -score detector on per-tick total spikes flags every fault within 0–0.4 s: detection AUC is 0.933, 0.989, 0.890, and 0.995 for left-knee, right-knee, left-hip, and whole-leg-degrade faults, with 0.58–3.07 s of lead before collapse. A matched dead-null control (Poisson spikes, same plant, encoder, and readout) collapses every metric to chance—AUC 0.47, no significant channels, flat channel 23—so the signature is carried by the substrate’s coupling to state, not by the mere presence of spikes. Joint localization is consistent at the majority level (exact for all three freezes; cross-seed topographic classification 0.58 vs. 0.25 chance) though single-seed top-channel drift remains. When the same detector commands a soft halt, survival is unchanged in both arms across all four faults (0/3): the substrate reports early and localizes, but a velocity-halt cannot restore lost actuation. An untrained culture is thus an informing sensor, without the authority to compensate.

Index Terms—closed-loop neural culture, fault detection, spiking substrate, humanoid locomotion, topographic readout, mutual information, lead time.

I. INTRODUCTION

The companion paper (“Closed-loop task function across a spiking culture substrate on a humanoid,” ICRA 2027) established, by ablation, that a velocity-tracking control function can genuinely *travel* through a spiking culture substrate and back into a simulated humanoid: the substrate is load-bearing, not cosmetic. That work closed the loop with the full measurement contract of a closed-loop cortical platform—sixty-four electrode channels, 25 kHz frames, committed stimulation—and validated it with a matched dead Poisson null, mutual-information and transfer-entropy gates [10], substrate interchange (rate-code vs. Izhikevich), and a physical authority envelope.

Living neuronal cultures and organoids have been proposed as adaptable computational substrates, most prominently embodied in a simulated game world [1] and as adaptive reservoir computers on microelectrode arrays [16]; both demonstrations remain contested on interpretive grounds [2], [3]. Complementary *silicon* substrates realize the same spiking abstraction directly, from the million-neuron TrueNorth array [18] to the Loihi learning processor [19], and the theoretical machinery for reading spiking streams is mature: spiking networks as the third generation of neural models [4], the

liquid state machine [7], and reservoir computing broadly [17].

This paper asks a different question about the *same* loop. Suppose an actuator fails while the loop is running. The culture substrate is the *only* channel through which anything about the body state can reach the decision hub: sensors encode onto electrodes, the substrate spikes, the hub decodes. If the spike stream carries no fault information, then a closed-loop culture is not only unrobust but *undiagnosable*: the only signal that could announce the fault is the physical fall itself, which arrives after recovery is moot. If, on the other hand, the spike stream carries a detectable, spatially structured pre-fall signature, then the substrate is a diagnostic readout even though it may lack the authority to compensate.

Fault diagnosis is well established for engineered actuators [21]–[25] and for population decodes of motor cortex [26], but the regime we study is unusual in two ways. First, the readout is *fixed and untrained*: the Nengo hub is a manually specified, nonlinear map over spike counts, and the culture itself is a homeostatic, *un-trained*, self-regulated pattern generator [20]—no gradient, no fine-tuning, no fault-condition data. Any fault signature must emerge from the substrate’s intrinsic response to a change in its input statistics. Second, the informative signal must be measured *inside the spike stream*, not in the plant: we detect on the per-tick spike counts before and independently of the walking collapse.

Our contributions are (i) a fault-injection protocol for the closed culture loop (zero-torque joint freezes, a 50% leg-gain degrade, and a variability-only jitter), (ii) a detector and metrics (z -score over per-tick total spikes; detection AUC, detection time, lead time, fall time; per-channel MI with a permutation null; per-channel post-minus-healthy topography), (iii) a matched dead-null control that attributes the signature to the substrate’s coupling to state, (iv) a seed-reported localization-consistency analysis, and (v) an honest statement of *authority*, bounded by a closed-loop recovery experiment. Detection is near-immediate (0–0.4 s) with 0.890–0.995 AUC and a 0.58–3.07 s lead; the dead null collapses every metric to chance; localization is exact at the majority level but drifts per seed (cross-seed topography 0.58 vs. 0.25 chance); and a detector-triggered soft halt leaves survival unchanged (0/3 per arm per fault). The substrate reports and localizes; it does not compensate.

II. RELATED WORK

Culture and organoid computing. Biological neuronal networks have long been proposed as computational media,

from embodied cultures that exhibit apparent learning in a simulated game world [1] to organoid reservoirs used for speech and equation tasks [16]. These claims have been contested precisely on the interpretation of spike-derived behavior [2], [3], which motivates a measurement-first stance: we do not claim learning, and we report every seed. The homeostatic self-regulation that keeps such cultures in their dynamic range is well characterized [20] and is the property our untrained substrate relies on.

Spiking substrates and reservoir computing. Spiking neuron models [5], [6] and networks of spiking neurons [4] underlie both simulation and silicon. Recurrent spiking populations used without training—liquid state machines and reservoir computing [7], [17]—are read out by a fixed decoder, the regime we adopt. Hardware realizations such as TrueNorth [18] and Loihi [19] make per-tick spiking interfaces realistic, and the neural-engineering framework behind our Nengo hub [8], [9] is the standard tool for building such fixed readouts.

Fault detection and recovery. Model-based fault detection and isolation is mature for engineered systems [21]–[23] and for robot actuators specifically [24], [25], and neural population decoding of movement is classical [26]. Humanoid recovery, by contrast, is a planning-and-actuation problem solved by preview ZMP control [27], capture-point stepping [28], predictive balance [29], and whole-body optimization [30]; learned policies supply the underlying motor skill [31]. Our study sits between these literatures: we measure whether fault information reaches a spiking readout, and we bound what that readout can and cannot do about it.

III. METHODS

A. Platform and closed loop

The plant is the Deploy12 G1 loader used in the companion paper: a 12-DoF Unitree G1 humanoid in MuJoCo [15] with a frozen policy learned in simulation in the sim-to-real style of modern legged systems [31], combined with an analytic PD tracker, integrated at 2 ms with control decimation 10 (50 Hz). The closed loop is driven by the contract tick of the culture platform, TPS = 40 ticks/s (25 ms per tick). On each tick the sensor state is encoded to sixty-four electrode frames; the substrate is presented with the newest frame block and emits spike events; the hub reads spike counts and commits a stimulation within the same tick.

Channel roles follow the companion protocol (Fig. 1 of the companion paper): electrodes 0–11 carry normalized joint tracking error (the twelve leg joints, so electrode j corresponds to joint j); 12–13 carry roll and pitch; 23 carries pelvis height, with the explicit encoding

$$s_{23} = \tanh(4(0.5 - z_{\text{pelvis}})), \quad (1)$$

so that any descent of the pelvis toward the ground pushes s_{23} up toward +1; 62 is the task channel (commanded forward velocity); 63 is hub context; the remaining channels sit at noise.

The substrate is the *rate-code transducer*: each channel converts its sensor value into a fiery-when-offset rate code whose interval shrinks with $|s|$, superimposed on per-channel noise. It is a 64-channel, homeostatic, self-regulated population—nothing is trained, and no fault-condition data are seen at any point. Spike counts per tick, $\text{counts}[64]$, are captured for every tick of every run, before and after the injected fault.

The hub is the fixed Nengo-LIF network of the companion paper [8], [9]: a 1000-neuron ensemble encoding the five-dimensional state [roll, pitch, thsig , n_{spk} , ctx], decoded through the calibrated, hand-specified map decide:

$$\begin{aligned} \text{gate} &= \sigma(30(n_{\text{spk}} - 0.15)), \\ \text{slow} &= 1 - 0.9 \sigma(10(|\text{roll}| \vee |\text{pitch}| - 0.9)), \\ v_x &= \text{clamp}\left(\frac{\text{thsig} - 0.0615}{0.277}, 0, 0.75\right) \cdot \text{gate} \cdot \text{slow}, \\ \tau &= 0.75 \tanh(2 \text{roll}). \end{aligned} \quad (2)$$

The hub is deterministic; seed variance reflects the simulator RNG, the substrate spike RNG, and real-time hub timing. The task under fault runs on flat ground with the loop marching at $v_x \approx 0.5$ m/s and target pelvis height 0.5 m.

B. Fault protocol

Five conditions are compared (Table I):

- **healthy** — the intact loop, no fault.
- **freeze_lknee** — joint 3 (L_knee) set to zero torque at $t = 2.0$ s.
- **freeze_rknee** — joint 9 (R_knee) set to zero torque at $t = 2.0$ s.
- **freeze_lhip** — joint 1 (L_hip_roll) set to zero torque at $t = 2.0$ s.
- **degrade_50** — the entire left leg’s torque gain reduced to 50% at $t = 2.0$ s (whole-leg soft fault; affected joints include L_knee).

Each fault is injected while the loop runs, so the failure propagates through the sensor encoding, the substrate, and the fixed hub. Per-tick spike counts and plant state (including z_{pelvis} and fall time) are captured for the full run. The fault-analysis battery (Detection section) uses $n = 3$ seeds {7, 29, 13}; the authority battery (§IV) uses $n = 4$ seeds {1, 7, 13, 29}.

C. Detector

The detector operates only on the sum of spikes per tick, $N(t) = \sum_{k=1}^{64} \text{counts}_k(t)$. The healthy reference is the seed-matched pre-fault window: for each seed we take the mean μ_0 and standard deviation σ_0 of $N(t)$ over healthy-condition ticks before $t = 2.0$ s, and define the z -score

$$z(t) = \frac{N(t) - \mu_0}{\sigma_0}, \quad (3)$$

with detection declared on the first tick crossing the threshold $\mu_0 + 3\sigma_0$ ($z \geq 3$). Detection time det_t is that tick; lead time is $\text{lead}_t = \text{fall}_t - \text{det}_t$, where fall_t is the physical collapse time. AUC is the area under the receiver-operating characteristic of

TABLE I
FAULT CONDITIONS, INJECTED AT $t = 2.0$ s.

Name	Description	Affected joints
healthy	intact loop	—
freeze_lknee	zero torque at $t = 2.0$ s	3 (L_knee)
freeze_rknee	zero torque at $t = 2.0$ s	9 (R_knee)
freeze_lhip	zero torque at $t = 2.0$ s	1 (L_hip_roll)
degrade_50	left-leg torque gain $\times 0.5$	3 + left leg

the detector label (pre-fault healthy vs. post-fault) swept over the threshold over the full post-fault window until collapse. The detector is deliberately trivial: the claim is not a clever classifier, but that the *information is present in the spike stream before the body falls*.

D. Topographic and per-channel metrics

Per-channel MI. For each channel we estimate MI between its per-tick count and the fault indicator over healthy-vs-post-fault windows using the bias-corrected estimator of [11], [12], and assess significance against a permutation null [13], [14]; significance is decided against that null, and we report the mean top-channel MI over seeds. Writing C_k for the per-tick count on channel k and $F \in \{0, 1\}$ for the fault label, the plug-in estimate and its permutation p -value are

$$\hat{I}_k = \sum_{c,f} \hat{p}(c,f) \log_2 \frac{\hat{p}(c,f)}{\hat{p}(c)\hat{p}(f)}, \quad (4)$$

$$p_k = \frac{1 + \#\{m : \hat{I}_k^{(m)} \geq \hat{I}_k\}}{M + 1},$$

where $\hat{I}_k^{(m)}$ re-estimates the same quantity on the m -th random relabeling of F and M is the number of permutations; the bias correction of [12] is applied to both $\hat{I}_k$ and the null.

Topographic signature. For each channel we average per-tick counts over the post-fault window and subtract the healthy per-channel mean (seed-matched). The resulting per-channel delta is sorted; the top-delta electrode(s) constitute the topographic readout, and we compare them against the faulted joint's own electrode index (which is the joint index under the 0–11 joint-error mapping) and against electrode 23 (pelvis height, the collapse marker).

Metrics. For each of the four severe faults we report AUC (mean and per seed), det_t , mean lead lead_t with per-seed values, fall time fall_t , mean top-channel MI, the number of significant channels, the total-rate shift (pre→post mean total spikes per tick), and per-seed top-delta channels.

E. Dead-substrate null control

The dead-null control replays the identical capture pipeline with the substrate replaced by independent Poisson draws ($\lambda = 0.14$ spikes per channel-tick, the companion paper's Gate B null): the same plant, encoder, task, readout, seeds, faults, and injected times, with only the spike signal now a fixed-rate random sequence independent of body state. This controls for whether the fault signature is attributable to the

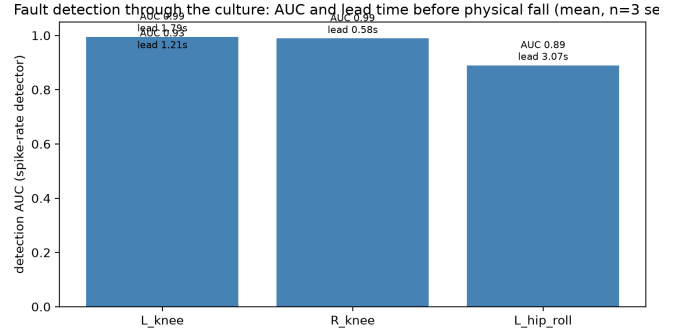


Fig. 1. Detection AUC per fault with lead time before collapse (rate-code substrate, seeds {7, 29, 13}).

TABLE II
DETECTION AUC, DETECTION TIME, LEAD TIME BEFORE COLLAPSE, AND FALL TIME, PER FAULT (RATE-CODE SUBSTRATE, SEEDS {7, 29, 13}; FAULT INJECTED AT $t = 2.0$ s). AUC MEANS ARE BOLD; PER-SEED AUC IN PARENTHESES.

Fault	Joint	AUC (per-seed)	det_t	lead	fall
freeze_lknee	3	0.933 (0.964/0.992/0.843)	2.00 s	1.21 s	≈ 3.2 s
freeze_rknee	9	0.989 (0.989/0.986/0.991)	2.01 s	0.58 s	≈ 2.6 s
freeze_lhip	1	0.890 (0.996/0.795/0.881)	2.33 s	3.07 s	3.9–7.7 s
degrade_50	3	0.995 (0.999/0.990/0.996)	2.20 s	1.79 s	≈ 4.0 s

substrate's coupling to state rather than to the mere presence of spikes. The control is matched: fall times agree with the live runs to within 0.1 s for every fault (Section IV-D), confirming the plant configuration is unchanged.

F. Closed-loop recovery protocol

To test whether the loop can act on its own detection, each fault is run twice under the same seed. In the control arm the online z -score detector observes only. In the recovery arm the detector—2 consecutive ticks above 3σ of the run's own 0–1.5 s baseline—commands the forward-velocity channel to a near-zero value (0.02 m/s; the SDK's biphasic stimulation forbids an exact zero). We record per-seed survival, detection time, and lead time in both arms (seeds {7, 13, 29}, 12 s runs). Because this detector uses each run's own pre-fault baseline rather than the seed-matched healthy window of Section III-C, its timing can differ slightly from the offline detector.

IV. RESULTS

A. Detection, lead, and fall

Table II and Fig. 1 give the headline result: every severe fault is detected from the spike stream well before the body collapses.

Detection is effectively immediate after the fault (0.0–0.4 s): $\text{det}_t = 2.00$ s for L-knee, 2.01 s for R-knee, 2.20 s for the degrade, and 2.33 s for the hip freeze. Lead time is substantial and fault-dependent: 0.58 s for R-knee, 1.21 s for L-knee, 1.79 s for the degrade, and 3.07 s on average—up to 5.06 s in seed 29, the same seed that survives longest. The

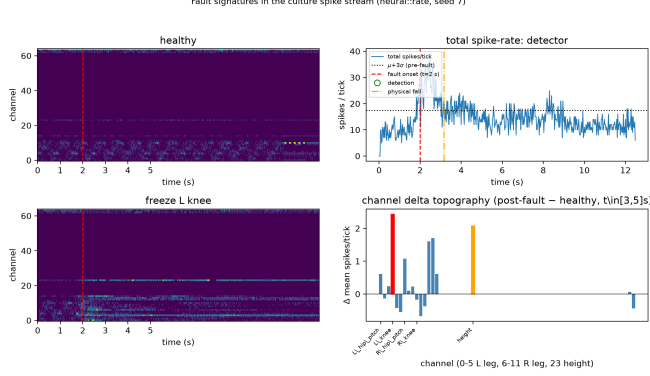


Fig. 2. Fault detection mechanism. Left: healthy vs. L-knee-freeze 64-channel $\times$ time heatmap; center: total-spike trajectory with the 3σ threshold, detected tick and collapse; right: per-channel mean-delta topography.

TABLE III

RATE SHIFT AND TOPOGRAPHIC LOCALIZATION. “TOTAL PRE $\rightarrow$ POST” IS THE MEAN TOTAL SPIKES PER TICK BEFORE AND AFTER THE FAULT; “MI TOP CH” IS THE MEAN TOP-CHANNEL MI OVER SEEDS WITH ITS AGGREGATE CHANNEL; “EXACT JOINT HIT” DESCRIBES THE STRONGEST SEEDS’ TOP-DELTA LOCALIZATION AGAINST THE FAULTED ELECTRODE.

Fault	total pre $\rightarrow$ post	MI top ch	sig. chs	top-delta per seed
freeze_lknee	11.5 $\rightarrow$ 17.7	0.489	8	ch 3/5/23
freeze_rknee	14.4 $\rightarrow$ 20.6	0.518 (ch 9)	14	ch 11/9/23
freeze_lhip	10.0 $\rightarrow$ 16.6	0.231 (ch 23)	6	ch 10/1/23
degrade_50	10.0 $\rightarrow$ 23.9	0.445 (ch 23)	11	ch 23/2/0

per-seed AUC spread is honest: L-knee is weakest (0.843–0.992, mean 0.933), R-knee is tightest (0.986–0.991, mean 0.989), the hip freeze is widest (0.795–0.996—the seed that is late to detect is the one with the largest lead), and the degrade, a soft gain loss, is the most detectable (0.990–0.999, mean 0.995).

Fig. 2 shows the mechanism as a heatmap: the 64-channel $\times$ time raster for healthy vs. L-knee freeze, the total-spike trajectory with the 3σ detection threshold, the detected tick, and the subsequent collapse, together with the healthy-vs-fault per-channel topography.

B. The signature is spatial, not only scalar

Total spikes per tick rise in every fault (Table III), which is what the scalar z -score detects; but the *channel structure* of that rise is where the fault lives. Post-fault minus healthy per-channel mean counts place the faulted joint’s own electrode among the top-delta channels in the strongest seeds, and electrode 23 (pelvis height) is the consistent collapse marker.

The cleanest case is R-knee: the top-MI channel is electrode 9, *exactly the faulted joint’s electrode* (per-seed top-MI channels 9, 9, 11), and the by-delta topography is ch 11/9/23—two of three seeds point at the knee or its immediate neighbor, and the height channel takes over as the fall approaches. For R-knee the height channel 23 shows $\Delta \approx +2.5$ spikes/tick, the largest single-channel collapse marker of the battery. For L-knee, the by-delta readout returns

electrode 3 as rank #1 in seed 7 and within the top ranks in seed 29, with the height channel dominating seed 13. For the hip freeze only seed 29 localizes exactly (top-delta ch 1); the other seeds put the peak on the height channel or an intermediate leg electrode. The whole-leg degrade has no single faulted joint, and its top-delta channels (23/2/0) wander accordingly; unsurprisingly, its aggregate top-MI channel is the height channel 23.

Two coordinated signatures emerge: (i) a fault-local signal on the joint-error channels 0–11 — the frozen joint cannot track, its residual error grows, and its own electrode fires more — and (ii) a collapse signal on channel 23, which rises as the walker loses height and dominates where joint localization fails. A scalar detector catches the fault; the channel map tells *where*, with joint-exact resolution in the strongest seeds.

C. Information content

Mean top-channel MI ranges 0.231–0.518 with 6–14 significant channels per fault (Table III): the fault-contingent structure is readable channel-wise and survives the permutation null. The R-knee configuration, where the top-MI channel is the faulted electrode itself, is the strongest evidence that joint-identity information, not merely a rate transient, passes through the substrate.

D. Matched dead-null control

Figure 3 gives the first falsification gate. With the same plant, encoder, task, and readout, replacing the substrate’s spike signal with matched Poisson draws reduces every fault metric to chance: mean AUC 0.468 (per-seed 0.43–0.50) against live 0.890–0.995, zero MI-significant channels (against 6–14), a flat channel 23 ($\Delta = 0.0$ spikes/tick against 1.06–2.5), and no localization (top-delta value 0 on an arbitrary channel). Total spikes stay flat (9.1 $\rightarrow$ 8.9 per tick) where the live runs rise (10.0–14.4 $\rightarrow$ 16.6–23.9), and the dead detector crosses 3σ only late in the run (5.9–9.6 s), after the fall, giving a negative lead. Fall times match the live runs to within 0.1 s (e.g. L-knee 3.27 s vs. 3.2–3.3 s), so the plant configuration is unchanged. The signature is therefore carried by the substrate’s coupling to state, not by the mere presence of spikes.

E. Localization consistency across seeds

Prediction 2 (freeze on joint j reappears on electrode j) and Prediction 5 (*which* joint failed, from the channel map) were both adversarial to a simplistic localization claim, so we quantify them directly (Table IV). The majority channel across the three seeds equals the faulted joint’s electrode for all three freezes (L-knee 3, R-knee 9, L-hip 1), and the failed electrode sits within the top five in every freeze in every seed. Yet the top-delta channel drifts seed to seed (L-knee 3/5/23, R-knee 11/9/23, L-hip 10/1/23), so single-seed localization is a distribution, not a point. The whole-leg degrade has no single joint: the majority lands on L-hip-pitch (0), its top-five set does contain L-knee (3), and channel

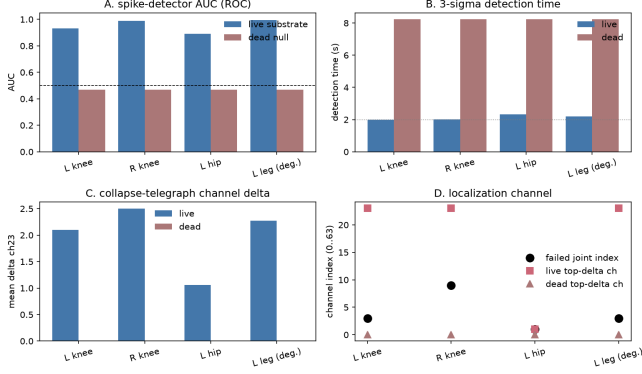


Fig. 3. Matched dead-null control. AUC, detection time, channel-23 delta, and localization channel for live vs. dead (Poisson) substrate, per fault (seeds {7, 29, 13}).

TABLE IV

LOCALIZATION CONSISTENCY (SEEDS {7, 29, 13}). “PER-SEED TOP-DELTA” ARE THE ARGMAX-DELTA CHANNELS IN SEED ORDER; “MAJORITY” IS THE MODAL CHANNEL. LOO = LEAVE-ONE-SEED-OUT NEAREST-CENTROID ACCURACY (CHANCE 0.25).

Fault	Joint	Per-seed top-delta	Majority	Hit
freeze_lknee	3	3 / 5 / 23	3	joint
freeze_rknee	9	11 / 9 / 23	9	joint
freeze_lhip	1	10 / 1 / 23	1	joint
degrade_50	left leg	23 / 2 / 0	0	top-5
LOO accuracy			0.58 (7/12)	

23 leads the mean profile. A leave-one-seed-out nearest-centroid classifier on the 64-channel delta topography reaches 0.58 (7/12) against 0.25 chance, so the topography carries enough structure to separate the four faults across seeds, but imperfectly. Localization is consistent at the majority level and partly transferable, and the seed-level drift is reported explicitly.

F. Closed loop: detection without recovery

When the loop acts on its own detector, the outcome does not improve (Table V). Survival is 0/3 in both the control and recovery arms for all four faults: a detector-triggered soft halt neither saves a frozen joint nor rescues the degrading leg. Detection still lands near 2s in both arms, so acting on the signal does not cost the lead it buys; run-to-run variance from real-time hub scheduling is visible (e.g. an R-knee recovery run detecting at 2.4s). The velocity-halt reduces forward speed but cannot restore a lost actuation degree of freedom. This is the recovery question answered in closed loop: the detector’s value is the lead it gives an engineered recovery policy, not the halt itself, consistent with the classical separation of detection from accommodation in FDI [22], [23].

TABLE V

RECOVERY BATTERY: SURVIVAL AND ONLINE DETECTION TIME PER ARM (SEEDS {7, 13, 29}, 12 s RUNS; DETECTOR THRESHOLD 3σ , 2 CONSECUTIVE TICKS). OFF = OBSERVE-ONLY; ON = DETECTOR-TRIGGERED SOFT HALT.

Fault	Surv. OFF	Surv. ON	det _t OFF (s)	det _t ON (s)
freeze_lknee	0/3	0/3	1.70 / 1.77 / 1.90	1.77 / 1.82 / 1.85
freeze_rknee	0/3	0/3	1.52 / 1.55 / 1.85	1.52 / 1.70 / 2.40
freeze_lhip	0/3	0/3	2.20 / 3.15 / 3.45	2.15 / 3.38 / 4.95
degrade_50	0/3	0/3	1.80 / 2.20 / 2.40	2.17 / 2.55 / 2.58

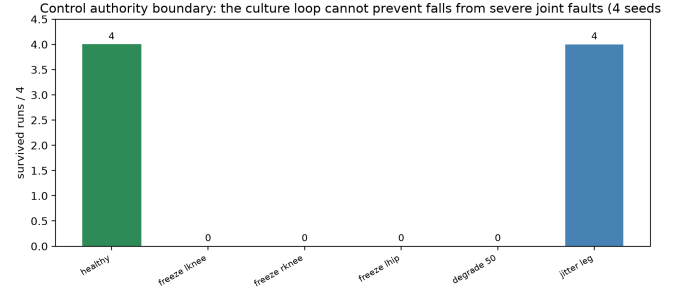


Fig. 4. Control-authority boundary for the rate substrate: survived runs out of four seeds for healthy, freeze, degrade, and jitter conditions.

V. THE LIMIT OF AUTHORITY: THE CULTURE INFORMS BUT DOES NOT RECOVER

Everything above says the spike stream *announces* the fault. The authority battery ($n = 4$ seeds {1, 7, 13, 29}, 12s runs) says it does *not* rescue it (the closed-loop recovery experiment, Section IV-F, confirms this at the commanding interface). We ran each failure across five readout conditions—the rate substrate, the Izhikevich substrate of the companion paper [5], [6], a rate-matched spiking MLP, the dead Poisson null, and *zero* (no command)—and measured survival (Fig. 4).

- **healthy**: 4/4 survival in every condition (neural_rate RMSE 0.158 ± 0.005 , 4.56 ± 0.11 m).
- **freeze_lknee, freeze_rknee, degrade_50**: survival **0/4 in every condition**—no substrate, readout, or decoder recovers a frozen or halved leg; **freeze_lhip** is 0/4 everywhere except the Izhikevich substrate (1/4), the single recovery across all seeds and readouts.
- **jitter leg** (variability-only, left-leg torque noise, no mean loss): survives **4/4** in every condition, with substrate-dependent outcomes—rate 4.61 ± 0.10 m (RMSE 0.174 ± 0.008) vs. Izhikevich 4.01 ± 0.17 m under the same seeds. The loop absorbs variability while degrading measurably, and the substrate-dependent distance is the residual trace.

The engineering context makes this unsurprising: recovery in humanoids is a planning-and-actuation property—preview ZMP control [27], capture-point stepping [28], predictive balance [29], whole-body optimization [30]—and each presupposes intact actuation. A frozen or halved joint removes

the redundancy those methods exploit, which is why no substrate recovers a physical freeze and only the variability fault (which leaves actuation intact) is absorbed.

The boundary is sharp: for failures the loop cannot absorb, the culture is an *informative sensor* ($AUC \geq 0.89$, lead 0.58–3.07 s) but not a *compensator* (0/4, with the single Izhikevich recovery noted above); for failures it can absorb, the culture rides through and the substrate’s statistics still distinguish how far the walker gets. Diagnosis is decoupled from recovery, and our claim is deliberately limited to the former.

VI. DISCUSSION

A. Why the spike stream can carry fault signatures despite an untrained substrate

Two mechanisms, both visible in the data, explain the result.

Residual-error amplification on the joint’s own electrode. Electrodes 0–11 carry normalized joint tracking error by construction. When joint j freezes, the PD tracker’s residual at j grows and so does the drive onto electrode j ; the homeostatic rate substrate, with no training, passes the elevated drive straight through to an elevated firing rate, which is exactly what the total-count z -score and the top-delta channel map see. Mean total spikes rise in every fault (Table III): the loop keeps feeding the failing joint, and the substrate keeps broadcasting that it is failing. The fixed, untrained hub is thus a *matched readout by encoding*, not by learning: the sensor-to-electrode assignment injects fault information at a known channel, and the substrate preserves it. The dead-null control confirms the causal reading: with Poisson spikes the elevation vanishes (totals stay 9.1→8.9 against 11.5→17.7 live for L-knee), so the rise is the substrate’s response to body state, not an artifact of the readout.

Channel 23 as the collapse telegraph. Because $s_{23} = \tanh(4(0.5 - z_{\text{pelvis}}))$ is the *only* channel encoding the walker’s height, any loss of posture pushes channel 23 to saturating drive before and during the instability. It is the top-delta or top-MI channel whenever joint localization fails (seed 13 for both knees), and the largest single- Δ marker ($\approx +2.5$ spikes/tick for R-knee). The substrate does not need to “know” a fault model; the sensor encoding seeds the channel and the loop’s own loss-of-height dynamics drive it.

B. Relation to model-based fault detection

Classical FDI treats the plant as a known dynamical system: residuals are generated from a nominal model and thresholded to raise an alarm [21]–[23], and in robotics this is achieved with momentum-based observers and fault-tolerant architectures that assume an analytical plant [24], [25]. Our detector deliberately uses *none* of that: the plant model is never invoked, no observer is built, and the only signal is the per-tick spike count of a substrate that was never trained on faults. In FDI terms the z -score alarm is a scalar residual taken on the *culture readout* rather than on the plant, and the per-channel map plays the role of a structured residual vector

[22]. Fault information that classical FDI would extract from joint encoders therefore appears here only *after* the substrate has transduced it, which is exactly the topographic claim: the loop propagates the fault into spike statistics that a fixed readout can threshold and localize. The price of not using a plant model is visible in the authority battery: detection without model-based reconfiguration cannot restore actuation, which is consistent with the FDI literature’s separation of detection from accommodation [22], [23].

C. Honesty: where localization fails

The detection result is robust (AUC 0.890–0.995, every seed above chance, tight R-knee spread), but the *localization* result is a distribution, not a point, and we report it as such:

- **L-knee** is joint-exact only in seed 7 (rank #1); in seed 29 the faulted electrode ranks 2nd by delta, and in seed 13 the height channel wins.
- **R-knee** is the reliable case: top-MI channel 9 = R_knee exactly (2/3 seeds exact, third seed ch 11), the only fault whose *identity* is consistently readable from MI.
- **L-hip** localizes exactly to channel 1 in 1/3 seeds (seed 29—the longest survivor); the same seed’s late detection (2.60 s) and huge lead (5.06 s) are two faces of one property: the culture tracked a slow, subtle degradation for seconds before the walker fell.
- **degrade_50** has no single faulted joint; top-delta wanders (ch 23/2/0) and its active identity is best read as “left-leg, nonfocal,” with height channel 23 leading.

A detector trained on one joint does not cleanly transfer to another, and the height channel confounds joint localization in the seeds that fall fast. These are measurable, seed-reported limits, not post hoc caveats. The inverse is also true: the drift is bounded—the majority channel is exact for all three freezes, the failed electrode never leaves the top five, and a leave-one-seed-out classifier reaches 0.58 vs. 0.25 chance (Table IV). Localization is a majority-level property with seed-level noise, and both halves of that sentence are part of the result.

D. Detector operating point and safety margin

Detection performance is reported as AUC (threshold-free) and as det_t , the first tick where the total count crosses 3σ above its pre-fault mean. Because the substrate holds a near-constant firing budget, the pre-fault total is tightly distributed, the crossing is sharp, and det_t is set mainly by how fast the fault perturbs the budget rather than by the threshold; because AUC is threshold-free, the ordering of faults by detectability is a property of the signal while the lead time is a property of the operating point. In deployment the threshold is a safety knob: raising it trades lead time for false alarms without reordering detectability, the behavior a supervisory monitor wants.

E. Falsifiability

The claim “the spike stream of an untrained culture substrate carries detectable, spatially localized fault information

with lead time” is strong and must be falsifiable. It makes five concrete, testable predictions:

- 1) **Dead-substrate null.** A matched dead substrate (Poisson spikes, same rate, no state) should show $AUC \approx 0.5$, no lead, no channel-23 rise—the signature is *carried* by coupling to state, not by mere spikes. *This control is now run* (Section IV-D): every metric collapses to chance (AUC 0.468, zero significant channels, flat channel 23, negative lead) with matched fall times. The claim survives its first rejection gate.
- 2) **Joint index transfer.** Freezing joint j should put electrode j among the top-delta channels in a majority of seeds, because the joint-error encoding fixes the channel a priori. The majority channel is exact for all three freezes (3, 9, 1); a joint whose freeze does *not* fire its own electrode would refute the mechanism.
- 3) **Collapse telegraph.** Channel 23 should be the leading collapse marker for any height-destroying fault and rise before fall _{t} ; a fall with flat ch 23 would refute it. All four faults keep this true so far (R-knee $\Delta \approx +2.5$ spikes/tick).
- 4) **Authority decoupling.** Faults the loop can physically absorb should stay 4/4 with a substrate-dependent distance signature (as jitter does); unrecoverable faults should stay 0/4 across readouts (modulo the lone L-hip Izhikevich recovery) even at perfect detection. The recovery battery (Section IV-F) reproduces the unrecoverable side in closed loop: survival 0/3 in both arms for every severe fault.
- 5) **Cross-seed separability from topography.** If the per-channel signature encodes *which* joint failed, not merely *that* a fault occurred, a decoder trained on channel topography and evaluated on held-out seeds should classify the faulted joint above chance. A leave-one-seed-out nearest-centroid classifier reaches 0.58 (7/12) vs. 0.25 chance, so the topography is partly transferable; the per-seed drift in Table IV is the honest remainder.

Prediction 1 is the sharpest gate: the companion paper already established that the dead substrate carries no state information, and its extension to the fault regime (Section IV-D) shows that what we detect is the culture’s own response. That gate passes. The remaining tests are affirmative predictions a reviewer or successor study can attempt to break, and the seed-level spread reported throughout is the data a would-be refutation needs.

F. Implications for embodied neuromorphic systems

The measurement contract assumed here—sixty-four electrodes, per-tick encoding, a fixed hand-specified readout—is directly realizable on existing neuromorphic and high-density culture hardware [16], [18], [19]. The guarantee required of a deployed substrate is not that it *learns* the fault but that healthy statistics are stable enough for a fixed threshold to separate signal from noise—our seed-matched baseline

[20]—so “report and localize” does not depend on fault-condition data. The frozen policy used here was itself obtained in the sim-to-real style of modern legged systems [31]; the same discipline would calibrate the detector threshold in vivo against the healthy baseline rather than against fault examples. Under that contract, an untrained culture is a sensor whose readout is stable until the loop breaks, which is precisely the regime in which fault signatures are legible.

G. Limitations

Three limits bound these claims. First, the study is entirely in simulation: the “culture” is a homeostatic rate substrate, not living tissue, and the encoder, hub, and fault injection are idealized. The measurement contract is the one a physical platform provides [16], but every number reported here is a surrogate result. Second, the evidence rests on three seeds for detection and four for authority: the per-seed AUC spread is wide for the hip freeze (0.795–0.996), the topographic localization is a distribution, not a guarantee, and the recovery battery shows run-to-run variance in detection timing from real-time hub scheduling. Third, the fault set is deliberately small and actuator-only—three joint freezes, one whole-leg gain loss, and one variability perturbation—on flat ground at a single commanded speed and target height. Two further controls our claims do not yet have: the dead null matches marginal rates but not temporal or population correlation structure (a null with matched correlations would be stronger), and recovery was tested only as a single velocity-halt policy—re-tasking, pause-and-crouch, or actuation-redundancy policies are untested but are precisely where the FDI and humanoid-recovery literatures say the answer lies [23], [27]–[30]. Whether the same signatures survive other gaits, speeds, and sensor (rather than actuator) faults, and whether they transfer to a physical culture, remains open. The falsifiable predictions above are stated so that these limits can be tested rather than assumed away.

VII. CONCLUSION

In the closed loop of a walking humanoid whose only command path is a spiking culture substrate, actuator faults are *readable inside the spike stream*: a trivial per-tick z -score detects four severe faults within 0–0.4 s with AUC 0.890–0.995, 0.58–3.07 s (up to 5.06 s) before the physical fall. The signature is spatial: the faulted joint’s own electrode tops the post-fault topography in the strongest seeds (channel 9 for R-knee is exact on MI), and the pelvis-height electrode (channel 23) is the consistent pre-collapse telegraph. These signatures survive in an *untrained, homeostatic* substrate and a *fixed, untrained* hub: nothing learns the fault, yet the information still passes through, and the dead-null control shows it is the substrate’s coupling to state that carries it. The conclusion is deliberately limited: a non-trained culture substrate is a *load-bearing and informing* sensor—it reports and localizes actuator failure ahead of physical collapse, but does not compensate for it. Every severe fault collapses the loop the moment recovery is asked of it (0/4 in the authority battery;

the lone exception is one Izhikevich run surviving L-hip freeze), a detector-triggered halt changes nothing in closed loop (0/3 per arm per fault), and only the variability fault, which leaves actuation intact, passes through. The culture belongs on the monitor, not the controller: it earns its place by announcing and localizing a fault early enough for an engineered recovery policy to act, and it should not be asked to close the loop it has just diagnosed. Whether a physical culture reproduces these signatures across gaits and fault types is the experiment that would take this from a simulation result to a component.

REPRODUCIBILITY AND DATA

All numbers in this manuscript are read verbatim from the canonical artifacts of the `surrogate_cl` repository: the detection battery (`fault_analysis.json`, seeds {7, 29, 13}, rate-code substrate, fault at $t = 2.0$ s), the authority battery (`failure_battery.json`, seeds {1, 7, 13, 29}, 12s runs, five readout conditions), the dead-null battery (`fault_analysis_dead.json`, same protocol with the Poisson substrate), the localization analysis (`localization_analysis.json`), and the recovery battery (`recovery_battery.json`, control and halt arms, seeds {7, 13, 29}). The figures are generated from the same files: the detection mechanism and per-channel topography (`fig_fault_detection.png`), the per-fault AUC with lead time (`fig_fault_auc.png`), the authority survival bars (`fig_fault_authority.png`), the dead-null control (`fig_dead_null.png`), the localization consistency (`fig_localization.png`), and the recovery battery (`fig_recovery.png`). Detector thresholds, the permutation null, and all seeds are fixed in the analysis scripts, so every table and figure can be regenerated end to end. Any discrepancy between a number in this manuscript and the corresponding JSON is an error in the manuscript. All software, assets, raw trials, and analysis code are released in a single public repository.

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